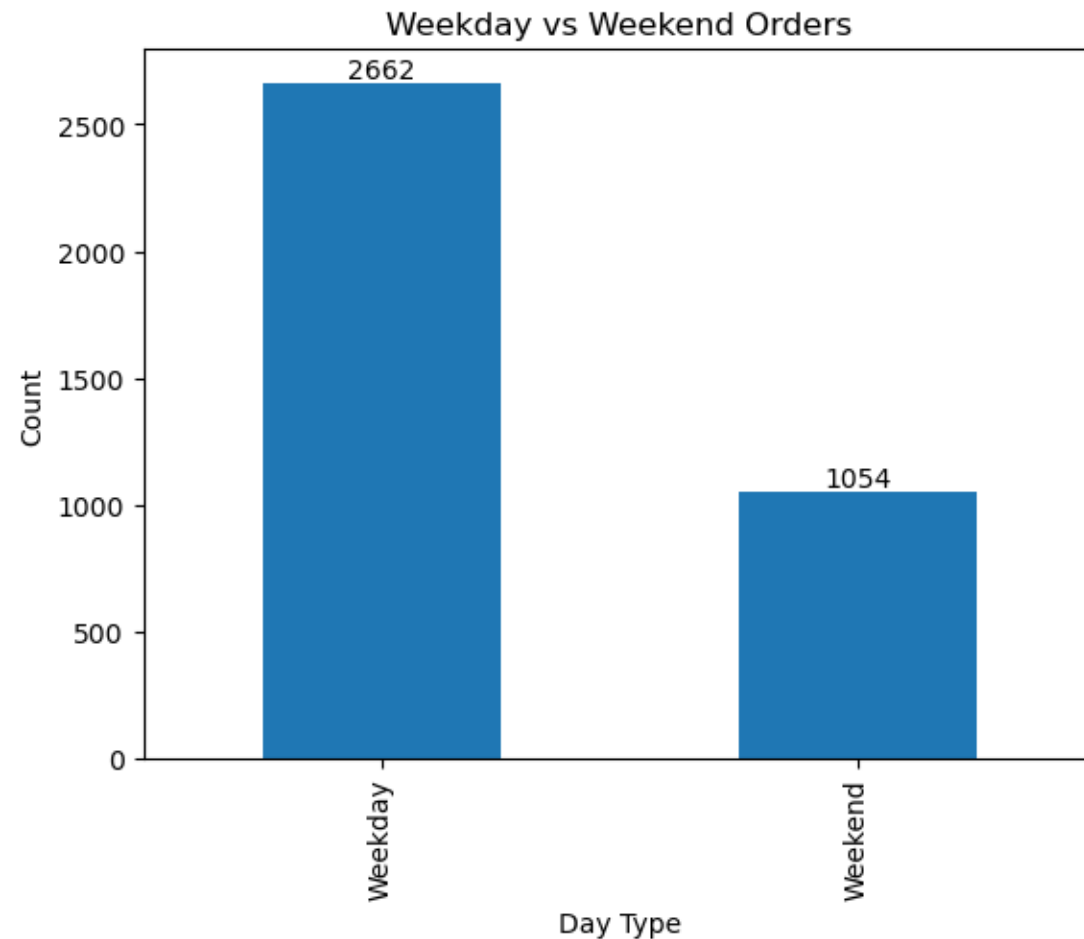


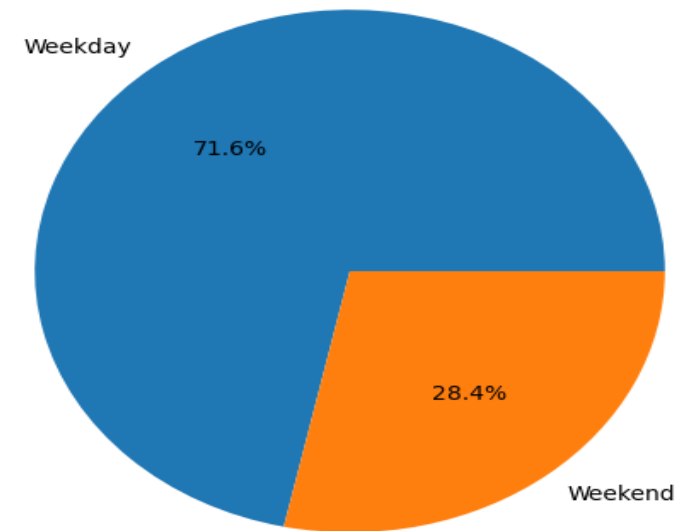
Kiddo Assignment

Vikrant Singh

Weekday vs Weekend Orders Analysis



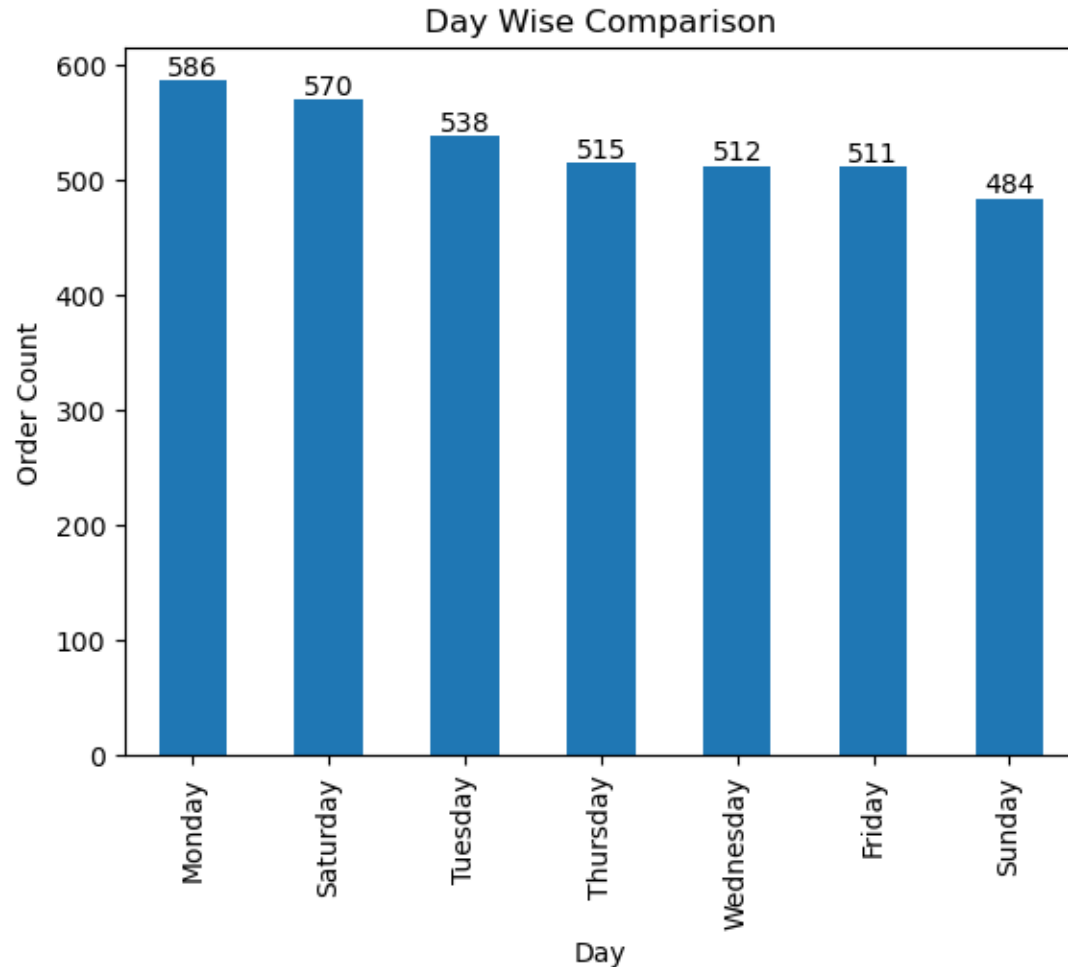
Weekday vs Weekend Orders Percentage Comparison



```
av_order_perday=df['Created At'].dt.date.value_counts().mean()  
print(f"Average Orders Per day: {round(av_order_perday,2)} ")
```

Average Orders Per day: 123.87

Weekday vs Weekend Orders Analysis



Key Observations

- Average Orders Per day: 123.87
- Demand is relatively balanced throughout the week.
- Resource planning should be demand based instead of day based.

Rider Deployment & Shift Planning

Observations

- Average Delivery Duration (Minutes): 41.81
- No of Orders Per Hour: 1.44

Assumptions

- Assuming 20 percent buffer time in delivery.
- Assuming 3 shifts (8 hours each)
- Riders Required for a shift is equal to maximum no of rides happening in any peak hours during that shift.

	Hour	Total_Orders	Avg_Orders_per_Hour	Rides Required	Shift
0	0	81	2.793103	3.0	Shift 1
1	1	30	1.764706	2.0	Shift 1
2	2	10	1.250000	2.0	Shift 1
3	3	7	1.000000	1.0	Shift 1
4	4	8	1.142857	1.0	Shift 1
5	5	4	1.000000	1.0	Shift 1
6	6	18	1.384615	2.0	Shift 1
7	7	52	2.000000	2.0	Shift 1
8	8	98	3.629630	4.0	Shift 2
9	9	140	4.666667	4.0	Shift 2
10	10	216	7.200000	7.0	Shift 2
11	11	256	8.533333	8.0	Shift 2
12	12	263	8.766667	8.0	Shift 2
13	13	238	7.933333	7.0	Shift 2
14	14	243	8.100000	7.0	Shift 2
15	15	216	7.200000	7.0	Shift 2
16	16	227	7.566667	7.0	Shift 3
17	17	235	7.833333	7.0	Shift 3
18	18	286	9.533333	8.0	Shift 3
19	19	283	9.433333	8.0	Shift 3
20	20	240	8.000000	7.0	Shift 3
21	21	229	7.633333	7.0	Shift 3
22	22	186	6.200000	6.0	Shift 3
23	23	150	5.000000	5.0	Shift 3

Peak Hour Analysis

Hour	Total_Orders	Avg_Orders_per_Hour
0	81	2.793103
1	30	1.764706
2	10	1.250000
3	7	1.000000
4	8	1.142857
5	4	1.000000
6	18	1.384615
7	52	2.000000
8	98	3.629630
9	140	4.666667
10	216	7.200000
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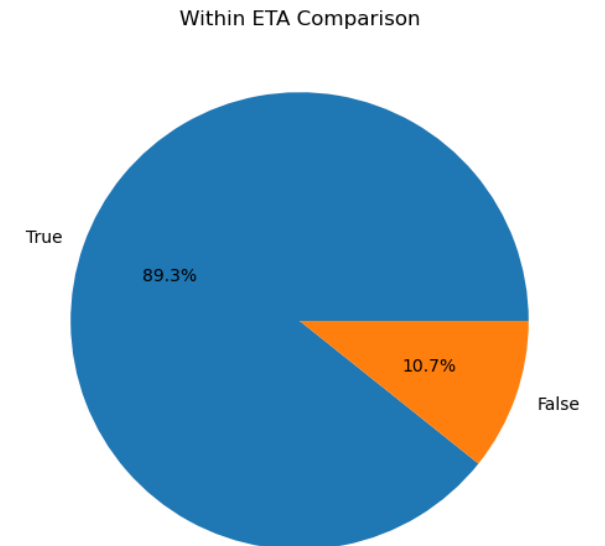
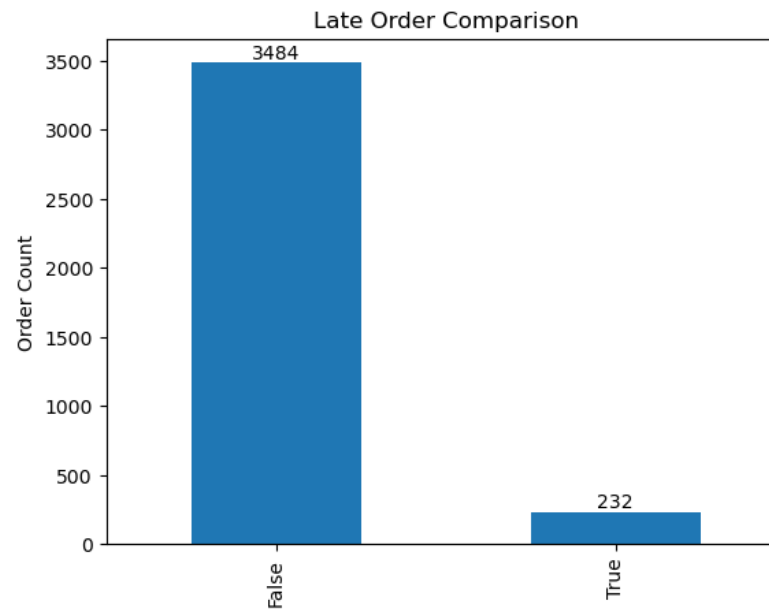
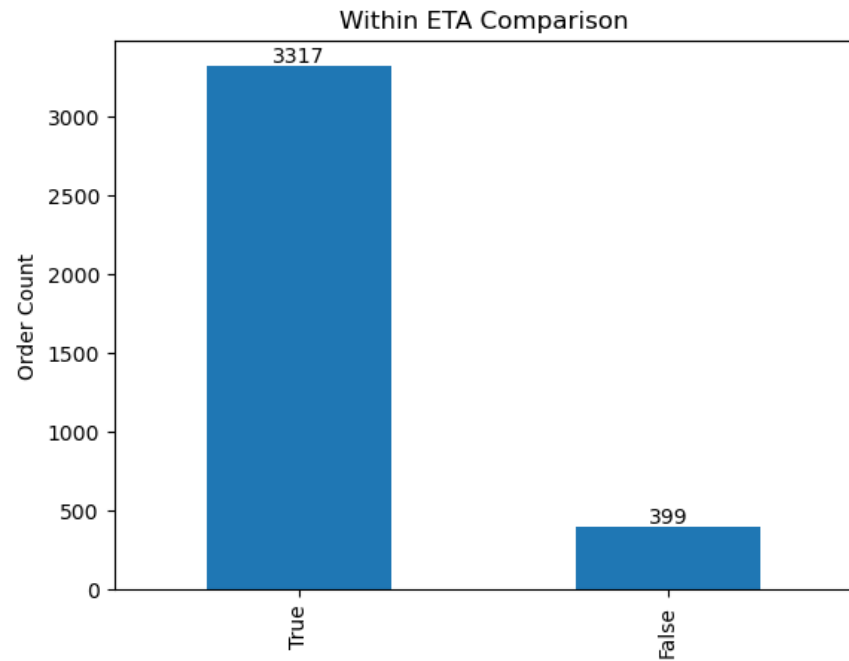
Rider Deployment & Shift Planning

Shift	Peak_Orders	Riders_Required
Shift 1	2.793103	3.0
Shift 2	8.766667	8.0
Shift 3	9.533333	8.0

ETA & Delivery Performance Analysis

Status	cancelled	delivered	exchanged	out_for_delivery	packed	packing	placed	reached_delivery	refunded	returned	rider_assigned
Order ID	786	7424	2	92	70	102	2	164	26	8	40

Average Delay (Minutes): 27.37



Rider Delay Analysis

Observations

- Average time from Out for Delivery to Delivered: 41.81 minutes

Top Delay Reason

- Item Not Available at Store
- Marked as Delivered Manually
- Distance > 25 km
- Item size

Recommendations

- Confirm inventory availability before rider assignment.
- Monitor orders exceeding **60 minutes** after Out for Delivery and trigger escalation alerts.
- Delays of **7-16 days** are operationally unrealistic for food or quick commerce deliveries. These records should be investigated as potential system timestamp errors, manual status updates, or exceptional cases, and may need to be excluded from KPI calculations to avoid skewing average delivery performance.

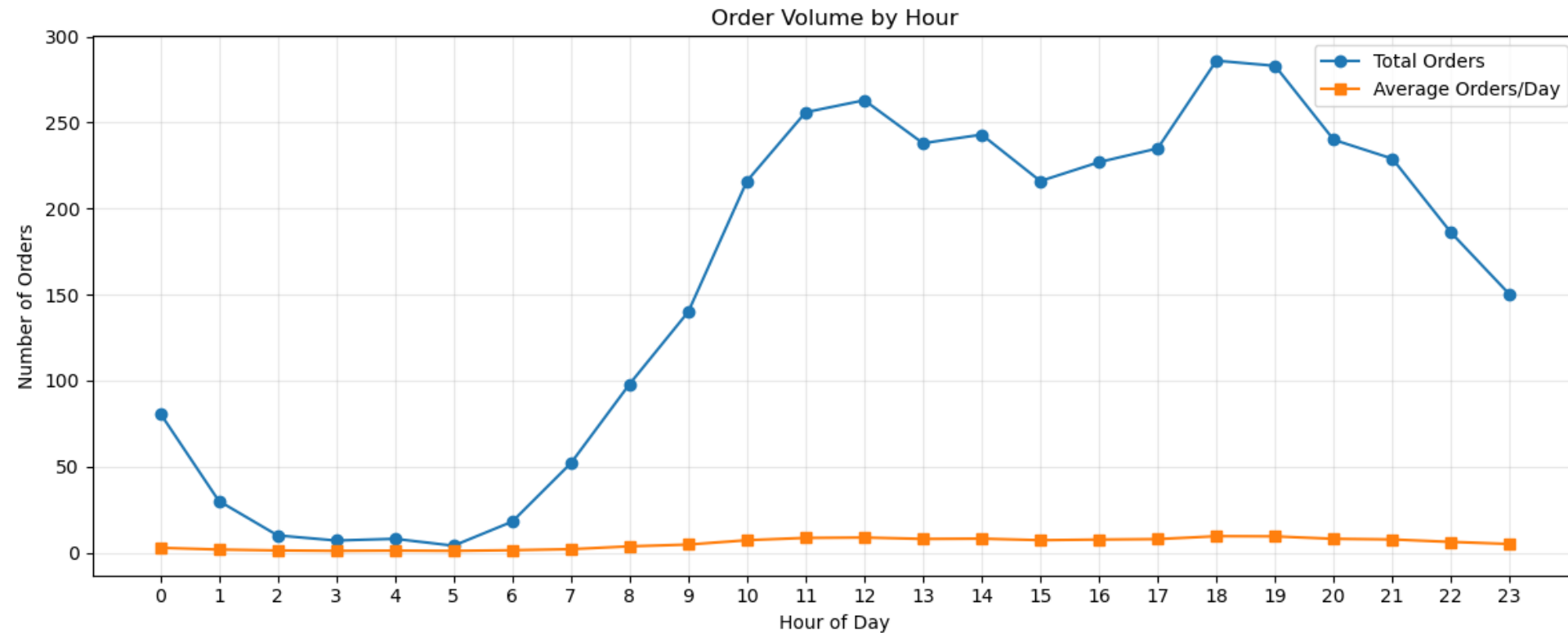
Order ID	Delivered Duration	delduration_min	Delay Reason
#14044	15 days 21:24:00	22884.0	bigger size not available
#16568	7 days 22:13:00	11413.0	Item NA at store
#17116	6 days 08:57:00	9177.0	size
#15388	2 days 23:14:00	4274.0	size
#17217	2 days 08:26:00	3386.0	Marked as delivered manually
#16559	2 days 03:38:00	3098.0	Marked as delivered manually
#14675	1 days 23:13:00	2833.0	scratches on the bottle
#14293	1 days 20:48:00	2688.0	size and quality issue
#17287	1 days 20:15:00	2655.0	Marked as delivered manually
#14281	1 days 19:56:00	2636.0	size issue , cx placed a new order
#18298	1 days 17:21:00	2481.0	Distance > 25km
#16397	1 days 04:15:00	1695.0	size
#14208	1 days 04:01:00	1681.0	-
#15989	0 days 23:26:00	1406.0	Marked as delivered manually
#16264	0 days 21:23:00	1283.0	Marked as delivered manually
#16022	0 days 21:23:00	1283.0	Marked as delivered manually
#14826	0 days 21:20:00	1280.0	cancel
#14563	0 days 21:18:00	1278.0	delivered
#14590	0 days 19:03:00	1143.0	Delivered
#14212	0 days 17:59:00	1079.0	-
#15306	0 days 17:35:00	1055.0	Marked as delivered manually
#15633	0 days 17:27:00	1047.0	Marked as delivered manually
#17427	0 days 16:54:00	1014.0	Marked as delivered manually
#18140	0 days 16:36:00	996.0	-
#18381	0 days 16:20:00	980.0	1st Item N/A at Store

Peak Hour Analysis

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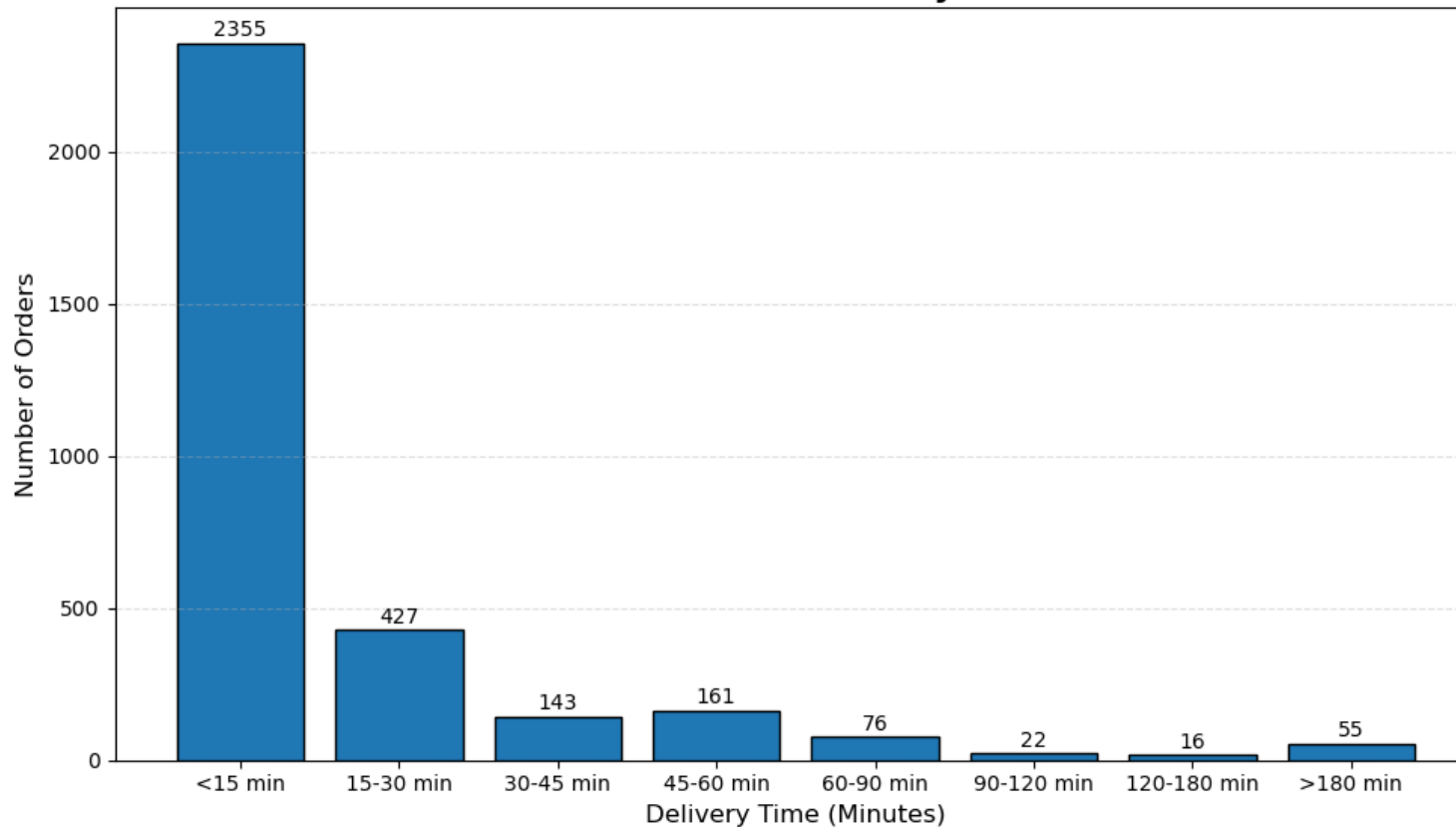
Peak Hour Analysis



Delivery Time Analysis

OBSERVATION OF SLOWEST AND
FASTEST ARE CALCULATED AFTER
REMOVING OUTLIER

Distribution of Delivery Times



Observations

- Average Delivery Duration (Minutes) : 41.81
- Median Delivery Time: 8.00 minutes
- Fastest Delivery: 1 Minute (Order Id #18377)
- Slowest Delivery: 174 min (Order Id #15516)

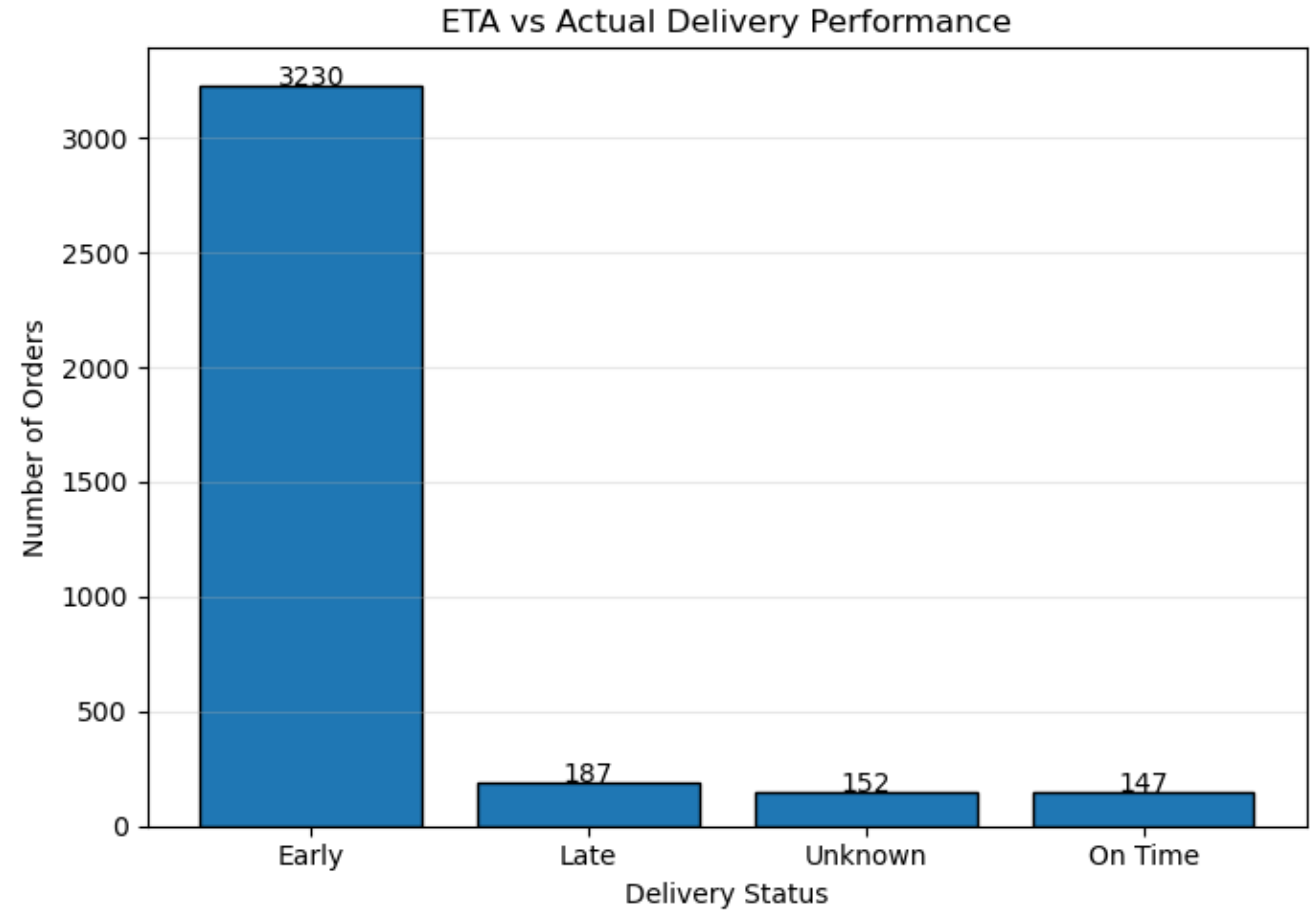
ETA Analysis

Observations

- Average ETA: 50.69 minutes
- Average ETA Variance: -8.96 minutes

Early Deliveries : 3317 (93.07%)
On-Time Deliveries : 15 (0.42%)
Late Deliveries : 232 (6.51%)

	Metric	Minutes
	Average ETA	50.69
	Average Actual Delivery	41.81



Thank you

[7088704043](#)

singh.vikrant9798@gmail.com

https://github.com/vinsin002/kiddo_assignment.git

<https://vinsin002.github.io/portfolio/>

You can find code files on my github. Link is above

This is not AI Generated PPT, I have just used template from Microsoft Templates.